

BASIL IHUOMA
MACHINE LEARNING ENGINEER

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SUMMARY

Machine Learning Engineer with 6+ years of experience building scalable, production grade ML systems end-to-end. Expertise in distributed training, ML platforms, feature stores, model lifecycle management, and decision-driven ML systems. Strong background in statistical modeling, time-series forecasting, and LLM systems, with a focus on reliability, cost-efficiency, and software engineering excellence.

PROFESSIONAL EXPERIENCE:

Machine Learning Engineer | LLM Trainer
Turing Enterprises Inc.

Jan. 2023 - Present
Palo Alto, California (Remote)

- Architected a **production ML platform** covering data pipelines, Feast + Redis feature store, distributed training, MLflow registry, Prefect orchestration, scalable inference, monitoring, and rollback, enabling reproducible end-to-end ML deployment.
- Designed **cost-aware, reliability-first ML systems**, optimizing Spark + Delta Lake pipelines, model serving, and monitoring for performance, fault tolerance, and cloud efficiency.
- Contributed to **LLM post-training workflows (RLHF)**, supporting data pipelines, evaluation loops, and scalable training infrastructure for alignment and model quality improvement.
- Built **decision-centric ML systems** that convert model outputs into optimization objectives, policy logic, and automated control loops for real-world operational use.
- Productionized **quantitative time-series forecasting systems** (Temporal Fusion Transformers, multi-horizon inference) with feature leakage prevention and live data/model drift monitoring.

Data | MLOps Engineer
Hamoye AI Labs.

May, 2020 - Dec. 2022
Hamilton, Bermuda (Remote)

- Architected and operated **distributed data & ML platforms on AWS** using Spark, Delta Lake, Kafka, and Kubernetes, enabling reliable large-scale batch and streaming pipelines.
- Designed **event-driven ingestion systems** (AWS Lambda → S3 → OpenSearch) for global datasets, reducing manual data handling by **70%+**.
- Implemented **Spark-on-Kubernetes workloads** with resource-aware scheduling, improving job reliability and cluster efficiency.
- Built **secure Delta Sharing architecture** for governed, cross-team data access.
- Established **software engineering standards**: Dockerized services, Git-based workflows, CI/CD pipelines, and unit/integration tests for data pipelines.

Data Scientist | Engineer
Hamoye AI Labs.

Feb, 2019 - April. 2020
Hamilton, Bermuda (Remote)

- Built **end-to-end ML workflows** for time-series forecasting (ARIMA, SARIMA, LSTM), including feature engineering, data validation, and reproducible experiment tracking for climate and energy datasets.
- Developed **Kubeflow pipelines** to orchestrate scalable ML training workflows, improving experiment reproducibility and reducing manual training overhead.
- Engineered **automated data acquisition pipelines** (Python scraping + cloud storage) to ingest remote sensing and climate raster data into cloud data lakes for downstream modeling.
- Designed **data models and preprocessing frameworks** for multivariate forecasting, improving model input consistency and reducing preprocessing errors across experiments.
- Standardized ML codebases with modular design, version control, and testing practices, improving reproducibility and reducing experiment-to-production friction.

Data Scientist | Analyst (Intern)
Alara Lagos.

Feb, 2018 - Jan. 2019
Victoria Island, Lagos, Nigeria.

- Built analytics pipelines and dashboards that informed merchandising decisions and reduced profit leakage by **8%**.
- Designed **customer and product segmentation models** that improved recommendation and inventory insights for business teams.
- Developed **marketing mix models** to measure channel performance, contributing to a **20% improvement in marketing ROI** within six months.
- Automated reporting workflows, improving operational efficiency by **7%**
- Delivered **production-grade dashboards and reports** (Tableau, Data Studio) with validated data sources and consistent metric definitions to improve executive decision-making.

TECHNICAL SKILLS

Languages: Python, SQL, C++

ML/AI: PyTorch, TensorFlow, Scikit-Learn, Hugging Face, LLM Post-Training (RLHF)

ML Systems & MLOps: MLflow, Prefect, Feast, Redis (online store), CI/CD, Experiment Tracking

Data Engineering: Apache Spark, Kafka, Delta Lake, PostgreSQL, Snowflake

Cloud & Infra: AWS, Kubernetes, Docker, Terraform

Model Serving & Monitoring: BentoML, FastAPI, Prometheus, Grafana, Evidently, Great Expectations

EDUCATION

Federal University of Technology, Owerri (FUTO)
B.Tech., Information Management Technology CGPA:

IMO State, Nigeria
4.2 / 5.0 Nov 2012 – Nov 2017

CERTIFICATIONS

Scientific Computing & Python for Data Science

WorldQuant University (2020)

Cloud DevOps Engineer (ALX-T Program)

Udacity (2022)

Mathematics for Machine Learning Track

Maths Academy

Coursework: Mathematical Foundations I–III, Linear Algebra, Probability & Statistics, Mathematics for ML